

Compiler and toolchain reference

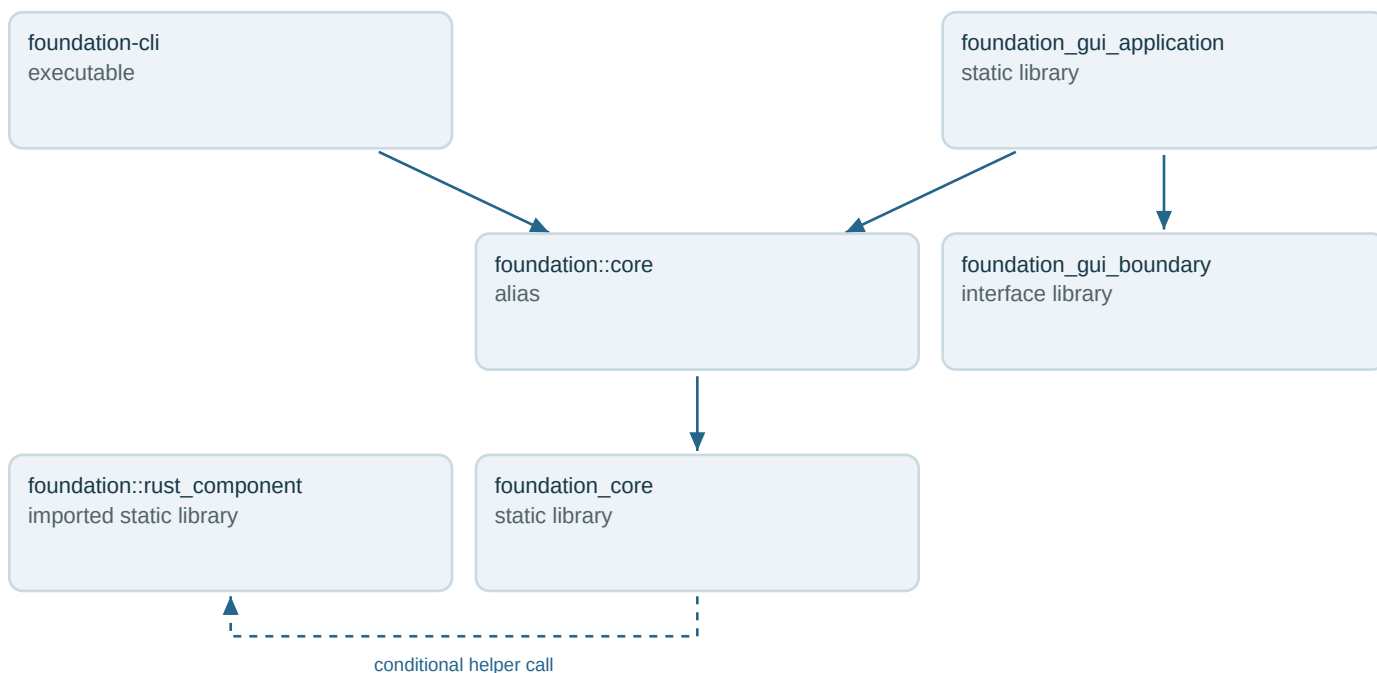
Source registration, link dependencies, configuration defaults, and the declarations behind them.

Captured 44 target records, 31 options/cache defaults, 31 toolchain assignments, and 1853 CMake commands.

This is a lexical navigation snapshot. CMake conditions, variables, helper bodies, and generator expressions are preserved as source text. The generator did not configure or compile the application. Target lists combine declarations across conditions and are not one selected configuration.

The snapshot is regenerated only on explicit request and may be out of date. Source links open the captured HTML explorer; every page links back to the edit-path guide.

1. Target directory and complete source/link details
2. Options and cache defaults
3. Compiler/toolchain assignments
4. Compiler, link, and dependency configuration commands
5. CMake presets and Cargo manifests
6. Build helpers and command inventory
7. Recorded limitations and warnings



Selected map: 6 of 44 target records. Solid arrows follow captured literal link/alias declarations. A dashed Rust arrow, when present, marks the observed conditional `foundation_rust_component(core)` helper call. Click a box for its complete target record.

1. Target directory

Every captured target record is listed below, including aliases, imported targets, helper declarations, and context-dependent duplicates. Click its name to inspect the complete source/link arguments and declaration context.

Target / kind	Declaration	Source / link entries
foundation_core static library	CMakeLists.txt:58	2 sources / 0 links
foundation::core alias	CMakeLists.txt:59	0 sources / 1 links
foundation-cli executable	CMakeLists.txt:105	1 sources / 1 links
foundation_core_test executable	CMakeLists.txt:142	1 sources / 1 links
foundation_windows_arguments_test executable	CMakeLists.txt:163	1 sources / 0 links
foundation_rust_component_test executable	CMakeLists.txt:176	1 sources / 1 links
foundation_text_status_test executable	CMakeLists.txt:190	2 sources / 0 links
foundation::rust_component imported static library	cmake/FoundationConfig.cmake.in:5	0 sources / 0 links
foundation::rust_component imported static library	cmake/RustComponent.cmake:81	0 sources / 0 links
consumer executable	examples/consumer/CMakeLists.txt:4	1 sources / 1 links
foundation_gui_boundary interface library	gui/CMakeLists.txt:73	0 sources / 1 links
gui_boundary alias	gui/CMakeLists.txt:74	0 sources / 1 links
foundation_gui_application static library	gui/CMakeLists.txt:104	2 sources / 2 links
\${target} executable	gui/CMakeLists.txt:116	3 sources / 1 links
foundation-gui-web-wasm executable (foundation_gui_executable helper)	gui/CMakeLists.txt:306	1 sources / 0 links
foundation-gui-terminal executable (foundation_gui_executable helper)	gui/CMakeLists.txt:362	1 sources / 0 links
foundation-gui-framebuffer executable (foundation_gui_executable helper)	gui/CMakeLists.txt:363	1 sources / 0 links
foundation-gui-web executable (foundation_gui_executable helper)	gui/CMakeLists.txt:365	1 sources / 0 links
foundation_gui_ftk interface library	gui/CMakeLists.txt:374	0 sources / 2 links

Target / kind	Declaration	Source / link entries
foundation-gui-fltk executable (foundation_gui_executable helper)	gui/CMakeLists.txt:384	1 sources / 1 links
foundation_gui_sdl interface library	gui/CMakeLists.txt:389	0 sources / 3 links
foundation-gui-sdl executable (foundation_gui_executable helper)	gui/CMakeLists.txt:402	1 sources / 1 links
foundation-gui-rev executable (foundation_gui_executable helper)	gui/CMakeLists.txt:417	1 sources / 1 links
foundation_gui_sdl_touch_test executable	gui/CMakeLists.txt:422	1 sources / 2 links
foundation_gui_\${example}_test executable	gui/CMakeLists.txt:452	1 sources / 1 links
foundation_gui_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:458	1 sources / 0 links
foundation_gui_terminal_caret_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:460	1 sources / 0 links
foundation_gui_task_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:462	1 sources / 0 links
foundation_gui_task_native_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:465	1 sources / 0 links
foundation_gui_file_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:467	1 sources / 0 links
foundation_gui_bezel_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:470	1 sources / 0 links
foundation_gui_file_transfer_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:472	1 sources / 0 links
foundation_gui_no_socket_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:475	1 sources / 0 links
foundation_gui_parity_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:485	1 sources / 0 links
\${test_target} executable	gui/CMakeLists.txt:499	1 sources / 1 links
foundation_gui_\${backend}_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:540	1 sources / 2 links
\${supplier_target} executable	gui/CMakeLists.txt:557	1 sources / 3 links
foundation_gui_supplier_clipboard executable	gui/CMakeLists.txt:579	1 sources / 2 links
foundation_gui_sdl_test executable (foundation_gui_executable helper)	gui/CMakeLists.txt:602	1 sources / 1 links
host_plugin module library	tests/cxx_runtime/CMakeLists.txt:8	1 sources / 0 links

Target / kind	Declaration	Source / link entries
foundation_core static library	tests/cxx_runtime/CMakeLists.txt:13	2 sources / 0 links
runtime_probe executable	tests/cxx_runtime/CMakeLists.txt:29	1 sources / 2 links
foundation_rev_probe static library	tests/rev_style_probe/CMakeLists.txt:26	11 sources / 0 links
foundation_rev_style executable	tests/rev_style_probe/CMakeLists.txt:35	1 sources / 1 links

Target 1: foundation_core

static library | [CMakeLists.txt:58](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
add_library(foundation_core STATIC src/store.cpp src/text_validation.cpp)
```

Source arguments (all)	Link arguments (all)
src/store.cpp	
src/text_validation.cpp	

Target 2: foundation::core

alias | [CMakeLists.txt:59](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

Alias of: foundation_core

```
add_library(foundation::core ALIAS foundation_core)
```

Source arguments (all)	Link arguments (all)
	foundation_core

Target 3: foundation-cli

executable | [CMakeLists.txt:105](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
add_executable(foundation-cli src/main.cpp)
```

Source arguments (all)	Link arguments (all)
src/main.cpp	foundation::core

Link statement | [CMakeLists.txt:106](#)

Context: no enclosing condition/helper recorded.

```
target_link_libraries(foundation-cli PRIVATE foundation::core)
```

Target 4: foundation_core_test

[executable](#) | [CMakeLists.txt:142](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
  add_executable(foundation_core_test EXCLUDE_FROM_ALL tests/store_test.cpp)
```

Source arguments (all)	Link arguments (all)
tests/store_test.cpp	foundation::core

Link statement | [CMakeLists.txt:143](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
  target_link_libraries(foundation_core_test PRIVATE foundation::core)
```

Target 5: foundation_windows_arguments_test

[executable](#) | [CMakeLists.txt:163](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(WIN32 AND MSVC AND NOT CMAKE_CROSSCOMPILING)
  add_executable(foundation_windows_arguments_test EXCLUDE_FROM_ALL
    tests/cxx_runtime/windows_arguments.cpp)
```

Source arguments (all)	Link arguments (all)
tests/cxx_runtime/windows_arguments.cpp	

Target 6: foundation_rust_component_test

[executable](#) | [CMakeLists.txt:176](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
  add_executable(foundation_rust_component_test EXCLUDE_FROM_ALL tests/rust_component_test.cpp)
```

Source arguments (all)	Link arguments (all)
tests/rust_component_test.cpp	foundation::core

Link statement | [CMakeLists.txt:177](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
  target_link_libraries(foundation_rust_component_test PRIVATE foundation::core)
```

Target 7: foundation_text_status_test

[executable](#) | [CMakeLists.txt:190](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
  add_executable(foundation_text_status_test EXCLUDE_FROM_ALL tests/rust_component_status_test.cpp
    src/store.cpp)
```

Source arguments (all)	Link arguments (all)
tests/rust_component_status_test.cpp	
src/store.cpp	

Target 8: foundation::rust_component

imported static library | [cmake/FoundationConfig.cmake.in:5](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(Foundation_CORE_PROVIDER STREQUAL rust AND NOT TARGET foundation::rust_component)
- add_library(foundation::rust_component STATIC IMPORTED)

Source arguments (all)	Link arguments (all)
------------------------	----------------------

No source or link arguments were recorded for this target. Imported/interface declarations may acquire them through settings or helpers shown in the source.

Target 9: foundation::rust_component

imported static library | [cmake/RustComponent.cmake:81](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
- add_library(foundation::rust_component STATIC IMPORTED)

Source arguments (all)	Link arguments (all)
------------------------	----------------------

No source or link arguments were recorded for this target. Imported/interface declarations may acquire them through settings or helpers shown in the source.

Target 10: consumer

executable | [examples/consumer/CMakeLists.txt:4](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

- add_executable(consumer main.cpp)

Source arguments (all)	Link arguments (all)
------------------------	----------------------

main.cpp

foundation::core

Link statement | [examples/consumer/CMakeLists.txt:5](#)

Context: no enclosing condition/helper recorded.

- target_link_libraries(consumer PRIVATE foundation::core)

Target 11: foundation_gui_boundary

interface library | [gui/CMakeLists.txt:73](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

- add_library(foundation_gui_boundary INTERFACE)

Source arguments (all)	Link arguments (all)
------------------------	----------------------

Threads::Threads

Link statement | [gui/CMakeLists.txt:83](#)

Enclosing context (unevaluated):

1. if(NOT EMSCRIPTEN)

```
target_link_libraries(foundation_gui_boundary INTERFACE Threads::Threads)
```

Target 12: gui_boundary

alias | [gui/CMakeLists.txt:74](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

Alias of: foundation_gui_boundary

```
add_library(gui_boundary ALIAS foundation_gui_boundary)
```

Source arguments (all)	Link arguments (all)
	foundation_gui_boundary

Target 13: foundation_gui_application

static library | [gui/CMakeLists.txt:104](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_library(foundation_gui_application STATIC shared/application.cpp ${foundation_task_executor})
```

Source arguments (all)	Link arguments (all)
shared/application.cpp	foundation_gui_boundary
<code>\${foundation_task_executor}</code>	foundation::core

Link statement | [gui/CMakeLists.txt:106](#)

Context: no enclosing condition/helper recorded.

```
target_link_libraries(foundation_gui_application PUBLIC foundation_gui_boundary foundation::core)
```

Target 14: \${target}

executable | [gui/CMakeLists.txt:116](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(${target} "${CMAKE_CURRENT_SOURCE_DIR}/${source}")
```

Source arguments (all)	Link arguments (all)
<code>\${CMAKE_CURRENT_SOURCE_DIR}/\${source}</code>	foundation_gui_application
<code>\${CMAKE_CURRENT_SOURCE_DIR}/hosts/windows_gui_entry.cpp</code>	
<code>\${CMAKE_CURRENT_SOURCE_DIR}/hosts/windows_gui.rc</code>	

Additional source statement | [gui/CMakeLists.txt:125](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)

2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)

```
target_sources(${target} PRIVATE "${CMAKE_CURRENT_SOURCE_DIR}/hosts/windows_gui_entry.cpp")
```

Additional source statement | [gui/CMakeLists.txt:132](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)

2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)

```
3. if(MSVC) -> else()
    target_sources(${target} PRIVATE "${CMAKE_CURRENT_SOURCE_DIR}/hosts/windows_gui.rc")
```

Link statement | [gui/CMakeLists.txt:135](#)

Enclosing context (unevaluated):

```
1. function(foundation_gui_executable target source)
    target_link_libraries(${target} PRIVATE foundation_gui_application)
```

Target 15: foundation-gui-web-wasm

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:306](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    foundation_gui_executable(foundation-gui-web-wasm hosts/wasm_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/wasm_main.cpp	

Target 16: foundation-gui-terminal

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:362](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
foundation_gui_executable(foundation-gui-terminal hosts/terminal_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/terminal_main.cpp	

Target 17: foundation-gui-framebuffer

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:363](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
foundation_gui_executable(foundation-gui-framebuffer hosts/framebuffer_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/framebuffer_main.cpp	

Target 18: foundation-gui-web

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:365](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_WEB)
    foundation_gui_executable(foundation-gui-web hosts/web_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/web_main.cpp	

Target 19: foundation_gui_fltk

interface library | [gui/CMakeLists.txt:374](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_library(foundation_gui_fltk INTERFACE)
```

Source arguments (all)	Link arguments (all)
	\${FLTK_LIBRARIES}
	X11::Xft

Link statement | [gui/CMakeLists.txt:377](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

```
target_link_libraries(foundation_gui_fltk INTERFACE ${FLTK_LIBRARIES})
```

Link statement | [gui/CMakeLists.txt:382](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

2. if(FOUNDATION_SDK_ROOT AND CMAKE_SYSTEM_NAME STREQUAL Linux)

```
target_link_libraries(foundation_gui_fltk INTERFACE X11::Xft)
```

Target 20: foundation-gui-fltk

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:384](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

```
foundation_gui_executable(foundation-gui-fltk hosts/ftk_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/ftk_main.cpp	foundation_gui_fltk

Link statement | [gui/CMakeLists.txt:385](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

```
target_link_libraries(foundation-gui-fltk PRIVATE foundation_gui_fltk)
```

Target 21: foundation_gui_sdl

interface library | [gui/CMakeLists.txt:389](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_SDL)

```
add_library(foundation_gui_sdl INTERFACE)
```

Source arguments (all)	Link arguments (all)
	SDL2::SDL2main
	SDL2::SDL2-static
	SDL2::SDL2

[Link statement](#) | [gui/CMakeLists.txt:391](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_SDL)
2. if(TARGET SDL2::SDL2main)
    target_link_libraries(foundation_gui_sdl INTERFACE SDL2::SDL2main)
```

[Link statement](#) | [gui/CMakeLists.txt:397](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_SDL)
2. if(FOUNDATION_PORTABLE)
    target_link_libraries(foundation_gui_sdl INTERFACE SDL2::SDL2-static)
```

[Link statement](#) | [gui/CMakeLists.txt:399](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_SDL)
2. if(FOUNDATION_PORTABLE) -> else()
    target_link_libraries(foundation_gui_sdl INTERFACE SDL2::SDL2)
```

Target 22: foundation-gui-sdl

[executable \(foundation_gui_executable helper\)](#) | [gui/CMakeLists.txt:402](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_SDL)
    foundation_gui_executable(foundation-gui-sdl hosts/sdl_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/sdl_main.cpp	foundation_gui_sdl

[Link statement](#) | [gui/CMakeLists.txt:403](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_SDL)
    target_link_libraries(foundation-gui-sdl PRIVATE foundation_gui_sdl)
```

Target 23: foundation-gui-rev

[executable \(foundation_gui_executable helper\)](#) | [gui/CMakeLists.txt:417](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_REV)
    foundation_gui_executable(foundation-gui-rev hosts/rev_main.cpp)
```

Source arguments (all)	Link arguments (all)
hosts/rev_main.cpp	gui_rev_adapter

[Link statement](#) | [gui/CMakeLists.txt:418](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_REV)
    target_link_libraries(foundation-gui-rev PRIVATE gui_rev_adapter)
```

Target 24: foundation_gui_sdl_touch_test

executable | [gui/CMakeLists.txt:422](#) | [HTML target view](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_SDL)

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(foundation_gui_sdl_touch_test "${CMAKE_SOURCE_DIR}/tests/gui_sdl_touch.cpp")
```

Source arguments (all)	Link arguments (all)
<code>\${CMAKE_SOURCE_DIR}/tests/gui_sdl_touch.cpp</code>	<code>foundation_gui_application</code>
	<code>foundation_gui_sdl</code>

Link statement | [gui/CMakeLists.txt:423](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_SDL)

```
target_link_libraries(foundation_gui_sdl_touch_test PRIVATE foundation_gui_application foundation_gui_sdl)
```

Target 25: foundation_gui_\${example}_test

executable | [gui/CMakeLists.txt:452](#) | [HTML target view](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. foreach(example touch embedding)

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(foundation_gui_${example}_test "${CMAKE_SOURCE_DIR}/tests/gui_${example}.cpp")
```

Source arguments (all)	Link arguments (all)
<code>\${CMAKE_SOURCE_DIR}/tests/gui_\${example}.cpp</code>	<code>foundation_gui_boundary</code>

Link statement | [gui/CMakeLists.txt:453](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. foreach(example touch embedding)

```
target_link_libraries(foundation_gui_${example}_test PRIVATE foundation_gui_boundary)
```

Target 26: foundation_gui_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:458](#) | [HTML target view](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)

```
foundation_gui_executable(foundation_gui_test tests/application_test.cpp)
```

Source arguments (all)	Link arguments (all)
<code>tests/application_test.cpp</code>	

Target 27: foundation_gui_terminal_caret_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:460](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
- foundation_gui_executable(foundation_gui_terminal_caret_test tests/terminal_caret_test.cpp)

Source arguments (all)	Link arguments (all)
tests/terminal_caret_test.cpp	

Target 28: foundation_gui_task_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:462](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
- foundation_gui_executable(foundation_gui_task_test tests/task_test.cpp)

Source arguments (all)	Link arguments (all)
tests/task_test.cpp	

Target 29: foundation_gui_task_native_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:465](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(NOT EMSCRIPTEN)
- foundation_gui_executable(foundation_gui_task_native_test tests/task_native_test.cpp)

Source arguments (all)	Link arguments (all)
tests/task_native_test.cpp	

Target 30: foundation_gui_file_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:467](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(NOT EMSCRIPTEN)
- foundation_gui_executable(foundation_gui_file_test tests/file_services_test.cpp)

Source arguments (all)	Link arguments (all)
tests/file_services_test.cpp	

Target 31: foundation_gui_bezel_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:470](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
- foundation_gui_executable(foundation_gui_bezel_test tests/bezel_test.cpp)

Source arguments (all)	Link arguments (all)
tests/bezel_test.cpp	

Target 32: foundation_gui_file_transfer_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:472](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
- foundation_gui_executable(foundation_gui_file_transfer_test tests/file_transfer_test.cpp)

Source arguments (all)	Link arguments (all)
tests/file_transfer_test.cpp	

Target 33: foundation_gui_no_socket_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:475](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(CMAKE_SYSTEM_NAME STREQUAL Linux)
- foundation_gui_executable(foundation_gui_no_socket_test tests/no_socket_test.cpp)

Source arguments (all)	Link arguments (all)
tests/no_socket_test.cpp	

Target 34: foundation_gui_parity_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:485](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
- foundation_gui_executable(foundation_gui_parity_test tests/parity_test.cpp)

Source arguments (all)	Link arguments (all)
tests/parity_test.cpp	

Target 35: \${test_target}

executable | [gui/CMakeLists.txt:499](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(test_name contract bitmap adapter layout runtime presentation extension interaction framebuffer terminal web)
- This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

add_executable(\${test_target} "\${test_source}")

Source arguments (all)	Link arguments (all)
\${test_source}	foundation_gui_boundary

Link statement | [gui/CMakeLists.txt:500](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(test_name contract bitmap adapter layout runtime presentation extension interaction framebuffer terminal web)
- target_link_libraries(\${test_target} PRIVATE foundation_gui_boundary)

Target 36: foundation_gui_\${backend}_test

executable (foundation_gui_executable helper) | gui/CMakeLists.txt:540 | [HTML target view](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_HOST_TESTS)
- 3. foreach(backend fltk rev)
- 4. if(FOUNDATION_GUI_\${upper})

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
foundation_gui_executable(foundation_gui_${backend}_test "tests/${backend}_test.cpp")
```

Source arguments (all)	Link arguments (all)
tests/\${backend}_test.cpp	gui_rev_adapter
	foundation_gui_fltk

Link statement | [gui/CMakeLists.txt:542](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_HOST_TESTS)
- 3. foreach(backend fltk rev)
- 4. if(FOUNDATION_GUI_\${upper})
- 5. if(backend STREQUAL rev)

```
target_link_libraries(foundation_gui_${backend}_test PRIVATE gui_rev_adapter)
```

Link statement | [gui/CMakeLists.txt:544](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_HOST_TESTS)
- 3. foreach(backend fltk rev)
- 4. if(FOUNDATION_GUI_\${upper})
- 5. if(backend STREQUAL rev) -> else()

```
target_link_libraries(foundation_gui_${backend}_test PRIVATE foundation_gui_fltk)
```

Target 37: \${supplier_target}

executable | [gui/CMakeLists.txt:557](#) | [HTML target view](#)

Enclosing context (unevaluated):

- 1. if(BUILD_TESTING)
- 2. if(FOUNDATION_GUI_HOST_TESTS)
- 3. foreach(backend fltk rev)
- 4. if(FOUNDATION_GUI_\${upper})

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(${supplier_target} "${supplier_source}")
```

Source arguments (all)	Link arguments (all)
\${supplier_source}	foundation_gui_boundary
	gui_rev_adapter

Source arguments (all)	Link arguments (all)
	foundation_gui_fltk

Link statement | [gui/CMakeLists.txt:559](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)

2. if(FOUNDATION_GUI_HOST_TESTS)

3. foreach(backend fltk rev)

4. if(FOUNDATION_GUI_\${upper})

target_link_libraries(\${supplier_target} PRIVATE foundation_gui_boundary)

Link statement | [gui/CMakeLists.txt:561](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)

2. if(FOUNDATION_GUI_HOST_TESTS)

3. foreach(backend fltk rev)

4. if(FOUNDATION_GUI_\${upper})

5. if(backend STREQUAL rev)

target_link_libraries(\${supplier_target} PRIVATE gui_rev_adapter)

Link statement | [gui/CMakeLists.txt:563](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)

2. if(FOUNDATION_GUI_HOST_TESTS)

3. foreach(backend fltk rev)

4. if(FOUNDATION_GUI_\${upper})

5. if(backend STREQUAL rev) -> else()

target_link_libraries(\${supplier_target} PRIVATE foundation_gui_fltk)

Target 38: foundation_gui_supplier_clipboard

executable | [gui/CMakeLists.txt:579](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)

2. if(FOUNDATION_GUI_HOST_TESTS)

3. if(FOUNDATION_GUI_REV)

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(foundation_gui_supplier_clipboard
"${CMAKE_CURRENT_BINARY_DIR}/tests/supplier_rev_clipboard_test.cpp")
```

Source arguments (all)	Link arguments (all)
\${CMAKE_CURRENT_BINARY_DIR}/tests/supplier_rev_clipboard_test.cpp	foundation_gui_boundary
	gui_rev_adapter

Link statement | [gui/CMakeLists.txt:580](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)

```
2. if(FOUNDATION_GUI_HOST_TESTS)
3. if(FOUNDATION_GUI_REV)
   target_link_libraries(foundation_gui_supplier_clipboard PRIVATE foundation_gui_boundary
   gui_rev_adapter)
```

Target 39: foundation_gui_sdl_test

executable (foundation_gui_executable helper) | [gui/CMakeLists.txt:602](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_HOST_TESTS)
3. if(FOUNDATION_GUI_SDL)
   foundation_gui_executable(foundation_gui_sdl_test tests/sdl_test.cpp)
```

Source arguments (all)	Link arguments (all)
tests/sdl_test.cpp	foundation_gui_sdl

Link statement | [gui/CMakeLists.txt:603](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_HOST_TESTS)
3. if(FOUNDATION_GUI_SDL)
   target_link_libraries(foundation_gui_sdl_test PRIVATE foundation_gui_sdl)
```

Target 40: host_plugin

module library | [tests/cxx_runtime/CMakeLists.txt:8](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
add_library(host_plugin MODULE plugin.cpp)
```

Source arguments (all)	Link arguments (all)
plugin.cpp	

Target 41: foundation_core

static library | [tests/cxx_runtime/CMakeLists.txt:13](#) | [HTML target view](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package)
```

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_library(foundation_core STATIC "${FOUNDATION_SOURCE}/src/store.cpp"
"${FOUNDATION_SOURCE}/src/text_validation.cpp")
```

Source arguments (all)	Link arguments (all)
`\${FOUNDATION_SOURCE}/src/store.cpp	
`\${FOUNDATION_SOURCE}/src/text_validation.cpp	

Target 42: runtime_probe

executable | [tests/cxx_runtime/CMakeLists.txt:29](#) | [HTML target view](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package) -> else()

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_executable(runtime_probe main.cpp)
```

Source arguments (all)	Link arguments (all)
main.cpp	\${CMAKE_DL_LIBS}
	foundation::core

Link statement | [tests/cxx_runtime/CMakeLists.txt:31](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package) -> else()

```
target_link_libraries(runtime_probe PRIVATE ${CMAKE_DL_LIBS})
```

Link statement | [tests/cxx_runtime/CMakeLists.txt:45](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package) -> else()

2. if(MODE STREQUAL protected) -> elseif(MODE STREQUAL installed)

```
target_link_libraries(runtime_probe PRIVATE foundation::core)
```

Target 43: foundation_rev_probe

static library | [tests/rev_style_probe/CMakeLists.txt:26](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

This record contains symbolic values. Resolve its variables and enclosing context in the source before registering another file.

```
add_library(foundation_rev_probe STATIC)
```

Source arguments (all)	Link arguments (all)
FILE_SET	
cxx_modules	
TYPE	
CXX_MODULES	
BASE_DIRS	
\${FOUNDATION_REV_SOURCE}/source	
FILES	
\${FOUNDATION_REV_SOURCE}/source/Core/DirtyFlag.ixx	
\${FOUNDATION_REV_SOURCE}/source/Core/Color.ixx	
\${FOUNDATION_REV_SOURCE}/source/Core/Resource.ixx	
\${FOUNDATION_REV_SOURCE}/source/Element/Style/Style.ixx	

Additional source statement | [tests/rev_style_probe/CMakeLists.txt:27](#)

Context: no enclosing condition/helper recorded.

```
target_sources(foundation_rev_probe PUBLIC FILE_SET cxx_modules TYPE CXX_MODULES
BASE_DIRS "${FOUNDATION_REV_SOURCE}/source" FILES
```

```
"${FOUNDATION_REV_SOURCE}/source/Core/DirtyFlag.ixx"  
"${FOUNDATION_REV_SOURCE}/source/Core/Color.ixx"  
"${FOUNDATION_REV_SOURCE}/source/Core/Resource.ixx"  
"${FOUNDATION_REV_SOURCE}/source/Element/Style/Style.ixx")
```

Target 44: foundation_rev_style

executable | [tests/rev_style_probe/CMakeLists.txt:35](#) | [HTML target view](#)

Context: no enclosing condition/helper recorded.

```
add_executable(foundation_rev_style style.cpp)
```

Source arguments (all)	Link arguments (all)
style.cpp	foundation_rev_probe

Link statement | [tests/rev_style_probe/CMakeLists.txt:36](#)

Context: no enclosing condition/helper recorded.

```
target_link_libraries(foundation_rev_style PRIVATE foundation_rev_probe)
```

2. Options and cache defaults

Defaults are values written in option()/set(... CACHE ...) declarations. They are not the effective values of a configured build.

FOUNDATION_SANITIZERS

CMakeLists.txt:13

Field	Captured value
default	OFF
description	Enable address and undefined-behavior checks

Context: no enclosing condition/helper recorded.

FOUNDATION_PORTABLE

CMakeLists.txt:14

Field	Captured value
default	OFF
description	Use a generic CPU floor and static language runtime

Context: no enclosing condition/helper recorded.

FOUNDATION_DISTRIBUTION_TESTS

CMakeLists.txt:20

Field	Captured value
default	OFF
description	Require local Debian repository tests and their tools

Context: no enclosing condition/helper recorded.

FOUNDATION_BUILD_GUI

CMakeLists.txt:28

Field	Captured value
default	OFF
description	Build optional pinned GUI boundary examples

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_SOURCE

CMakeLists.txt:29

Field	Captured value
default	(empty string)
cache_type	PATH
description	Explicit pinned GUI boundary checkout

Context: no enclosing condition/helper recorded.

FOUNDATION_CORE_PROVIDER

CMakeLists.txt:31

Field	Captured value
default	rust
cache_type	STRING
description	Text validation implementation: rust (default) or cpp

Context: no enclosing condition/helper recorded.

FOUNDATION_CONFIGURED_CORE_PROVIDER

CMakeLists.txt:40

Field	Captured value
default	\${FOUNDATION_CORE_PROVIDER}
cache_type	INTERNAL
description	Configured core provider

Context: no enclosing condition/helper recorded.

FOUNDATION_DEPENDENCY_RECIPE

CMakeLists.txt:309

Field	Captured value
default	native-unprepared
cache_type	STRING
description	Primary prepared dependency recipe

Context: no enclosing condition/helper recorded.

FOUNDATION_DEPENDENCY_RECIPES

CMakeLists.txt:310

Field	Captured value
default	(empty string)
cache_type	STRING
description	Complete retained recipe identities

Context: no enclosing condition/helper recorded.

FOUNDATION_RUNTIME_ROOTS

CMakeLists.txt:382

Field	Captured value
default	\${CMAKE_SYSROOT}
cache_type	STRING
description	Explicit target library roots for runtime collection

Enclosing context (unevaluated):

1. if(FOUNDATION_PORTABLE AND CMAKE_SYSTEM_NAME STREQUAL Linux)

FOUNDATION_DEPENDENCY_PREFIX

[cmake/BuildPolicy.cmake:3](#)

Field	Captured value
default	(empty string)
cache_type	PATH
description	Explicit verified rootless native development tree

Context: no enclosing condition/helper recorded.

FOUNDATION_CONFIGURED_DEPENDENCY_SHA256

[cmake/BuildPolicy.cmake:15](#)

Field	Captured value
default	\${prefix_sha256}
cache_type	INTERNAL
description	Configured native dependency bytes

Enclosing context (unevaluated):

1. if(FOUNDATION_DEPENDENCY_PREFIX)

FOUNDATION_CONFIGURED_DEPENDENCY_PREFIX

[cmake/BuildPolicy.cmake:27](#)

Field	Captured value
default	\${FOUNDATION_DEPENDENCY_PREFIX}
cache_type	INTERNAL
description	Configured native dependency input

Context: no enclosing condition/helper recorded.

FOUNDATION_COMPILER_CACHE

[cmake/BuildPolicy.cmake:28](#)

Field	Captured value
default	ON
description	Use an available ccache without requiring it

Context: no enclosing condition/helper recorded.

FOUNDATION_NOTICE_MANIFEST

[cmake/BuildPolicy.cmake:242](#)

Field	Captured value
default	(empty string)

Field	Captured value
cache_type	FILEPATH
description	Reviewed hash-bound notices for dependencies without distro ownership

Enclosing context (unevaluated):

1. function(foundation_finalize_build_policy)

FOUNDATION_WASM_PACKAGE

cmake/ImportWasm.cmake:5

Field	Captured value
default	(empty string)
cache_type	PATH
description	Complete prebuilt offline Wasm package directory

Context: no enclosing condition/helper recorded.

FOUNDATION_WASM_PACKAGE_SHA256

cmake/ImportWasm.cmake:6

Field	Captured value
default	(empty string)
cache_type	STRING
description	Exact web-manifest.json SHA-256

Context: no enclosing condition/helper recorded.

FOUNDATION_RUST_SDK_ROOT

cmake/RustComponent.cmake:3

Field	Captured value
default	(empty string)
cache_type	PATH
description	Verified retained Rust extension SDK

Context: no enclosing condition/helper recorded.

FOUNDATION_RUST_TARGET

cmake/RustComponent.cmake:4

Field	Captured value
default	(empty string)
cache_type	STRING
description	Explicit Rust target triple, or the matching platform default

Context: no enclosing condition/helper recorded.

FOUNDATION_CONFIGURED_RUST_IDENTITY

cmake/RustComponent.cmake:142

Field	Captured value
default	\${identity}
cache_type	INTERNAL
description	Configured Rust toolchain

Enclosing context (unevaluated):

- function(foundation_rust_component core)

FOUNDATION_HAS_TEST_PREREQUISITES

cmake/TestPrerequisites.cmake:87

Field	Captured value
default	ON
cache_type	INTERNAL
description	Complete generated test prerequisite inventory

Enclosing context (unevaluated):

- function(foundation_finalize_test_prerequisites)

FOUNDATION_NODE_EXECUTABLE

cmake/toolchains/sdk.cmake:34

Field	Captured value
default	\${FOUNDATION_SDK_ROOT}/node/bin/node
cache_type	FILEPATH
description	Retained target runtime

Enclosing context (unevaluated):

- if(_system STREQUAL Emscripten)

CMAKE_C_COMPILER

cmake/toolchains/sdk.cmake:76

Field	Captured value
default	\${_c_compiler}
cache_type	FILEPATH
description	Declared SDK C compiler

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_SOURCE

gui/CMakeLists.txt:5

Field	Captured value
default	(empty string)
cache_type	PATH
description	Verified gui-boundary source checkout or archive

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_INPUT_GROUP

[gui/CMakeLists.txt:6](#)

Field	Captured value
default	(empty string)
cache_type	PATH
description	Retained complete GUI input group

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_FLTK

[gui/CMakeLists.txt:25](#)

Field	Captured value
default	OFF
description	Build the FLTK native host

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_SDL

[gui/CMakeLists.txt:26](#)

Field	Captured value
default	OFF
description	Build the SDL framebuffer window host

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_REV

[gui/CMakeLists.txt:27](#)

Field	Captured value
default	OFF
description	Build the Rev native host

Context: no enclosing condition/helper recorded.

FOUNDATION_GUI_WEB

[gui/CMakeLists.txt:28](#)

Field	Captured value
default	ON

Field	Captured value
description	Build the hosted browser bridge
Context: no enclosing condition/helper recorded.	

FOUNDATION_GUI_HOST_TESTS

gui/CMakeLists.txt:29

Field	Captured value
default	OFF
description	Run PTY, loopback and native display checks
Context: no enclosing condition/helper recorded.	

GUI_REV_BUNDLED_DEPS

gui/CMakeLists.txt:412

Field	Captured value
default	ON
cache_type	BOOL
description	Build preserved local GLEW/FreeType sources

Enclosing context (unevaluated):

- 1. if(FOUNDATION_GUI_REV)
- 2. if(FOUNDATION_SDK_ROOT)

3. Compiler/toolchain assignments

All captured compiler/toolchain assignments are printed with their complete value tokens and enclosing source context. The selection includes CMAKE_* names and names containing COMPILER or TOOLCHAIN. Assignments in a toolchain/helper are declarations; no SDK or compiler is discovered or invoked here.

CMAKE_MSVC_RUNTIME_LIBRARY

CMakeLists.txt:26

Field	Captured value
Value token 1	MultiThreaded\$<\$<CONFIG:Debug>:Debug>

Enclosing context (unevaluated):

1. if(FOUNDATION_PORTABLE AND MSVC)

CMAKE_MSVC_RUNTIME_LIBRARY

CMakeLists.txt:55

Field	Captured value
Value token 1	\${_foundation_static_msvc_runtime}

Enclosing context (unevaluated):

1. if(MSVC AND FOUNDATION_CORE_PROVIDER STREQUAL rust)

CMAKE_\${language}_COMPILER_LAUNCHER

cmake/BuildPolicy.cmake:34

Field	Captured value
Value token 1	\${FOUNDATION_CCACHE_PROGRAM}

Enclosing context (unevaluated):

1. if(FOUNDATION_COMPILER_CACHE AND CMAKE_GENERATOR MATCHES Ninja|Makefiles)
2. if(FOUNDATION_CCACHE_PROGRAM)
3. foreach(language C CXX)
4. if(NOT CMAKE_\${language}_COMPILER_LAUNCHER)

FOUNDATION_RUST_COMPILER_VERSION

cmake/RustComponent.cmake:182

Field	Captured value
Value token 1	\${compiler_version}
Value token 2	PARENT_SCOPE

Enclosing context (unevaluated):

1. function(foundation_rust_component core)

FOUNDATION_RUST_COMPILER_SHA256

[cmake/RustComponent.cmake:183](#)

Field	Captured value
Value token 1	\${compiler_sha256}
Value token 2	PARENT_SCOPE

Enclosing context (unevaluated):

- 1. function(foundation_rust_component core)

CMAKE_TRY_COMPILE_TARGET_TYPE

[cmake/StaticCxxRuntime.cmake:12](#)

Field	Captured value
Value token 1	STATIC_LIBRARY

Enclosing context (unevaluated):

- 1. function(foundation_require_static_gnu_runtime)

CMAKE_REQUIRED_QUIET

[cmake/StaticCxxRuntime.cmake:13](#)

Field	Captured value
Value token 1	TRUE

Enclosing context (unevaluated):

- 1. function(foundation_require_static_gnu_runtime)

CMAKE_C_COMPILER

[cmake/toolchains/emscripten-rules.cmake:4](#)

Field	Captured value
Value token 1	\${FOUNDATION_SDK_ROOT}/bin/emcc

Context: no enclosing condition/helper recorded.

CMAKE_CXX_COMPILER

[cmake/toolchains/emscripten-rules.cmake:5](#)

Field	Captured value
Value token 1	\${FOUNDATION_SDK_ROOT}/bin/em++

Context: no enclosing condition/helper recorded.

CMAKE_AR

[cmake/toolchains/emscripten-rules.cmake:6](#)

Field	Captured value
Value token 1	\${FOUNDATION_SDK_ROOT}/bin/emar

Context: no enclosing condition/helper recorded.

CMAKE_RANLIB

cmake/toolchains/emscripten-rules.cmake:7

Field	Captured value
Value token 1	\${FOUNDATION_SDK_ROOT}/bin/emranlib

Context: no enclosing condition/helper recorded.

CMAKE_C_COMPILER_AR

cmake/toolchains/emscripten-rules.cmake:8

Field	Captured value
Value token 1	\${CMAKE_AR}

Context: no enclosing condition/helper recorded.

CMAKE_CXX_COMPILER_AR

cmake/toolchains/emscripten-rules.cmake:9

Field	Captured value
Value token 1	\${CMAKE_AR}

Context: no enclosing condition/helper recorded.

CMAKE_C_COMPILER_RANLIB

cmake/toolchains/emscripten-rules.cmake:10

Field	Captured value
Value token 1	\${CMAKE_RANLIB}

Context: no enclosing condition/helper recorded.

CMAKE_CXX_COMPILER_RANLIB

cmake/toolchains/emscripten-rules.cmake:11

Field	Captured value
Value token 1	\${CMAKE_RANLIB}

Context: no enclosing condition/helper recorded.

CMAKE_CXX_FLAGS_INIT

cmake/toolchains/sdk.cmake:37

Field	Captured value
Value token 1	-fexceptions

Enclosing context (unevaluated):

1. if(_system STREQUAL Emscripten)

CMAKE_EXE_LINKER_FLAGS_INIT

[cmake/toolchains/sdk.cmake:38](#)

Field	Captured value
Value token 1	-sASSERTIONS=1

Enclosing context (unevaluated):

1. if(`_system` STREQUAL Emscripten)

CMAKE_USER_MAKE_RULES_OVERRIDE

[cmake/toolchains/sdk.cmake:42](#)

Field	Captured value
Value token 1	<code>\${CMAKE_CURRENT_LIST_DIR}/emscripten-rules.cmake</code>

Enclosing context (unevaluated):

1. if(`_system` STREQUAL Emscripten)

CMAKE_SYSTEM_NAME

[cmake/toolchains/sdk.cmake:65](#)

Field	Captured value
Value token 1	<code>\${_system}</code>

Context: no enclosing condition/helper recorded.

CMAKE_SYSTEM_PROCESSOR

[cmake/toolchains/sdk.cmake:66](#)

Field	Captured value
Value token 1	<code>\${_processor}</code>

Context: no enclosing condition/helper recorded.

CMAKE_CXX_COMPILER

[cmake/toolchains/sdk.cmake:67](#)

Field	Captured value
Value token 1	<code>\${_cxx_compiler}</code>

Context: no enclosing condition/helper recorded.

CMAKE_C_COMPILER

[cmake/toolchains/sdk.cmake:76](#)

Field	Captured value
Value token 1	<code>\${_c_compiler}</code>
Value token 2	CACHE
Value token 3	FILEPATH
Value token 4	Declared SDK C compiler

Field	Captured value
Value token 5	FORCE
Context: no enclosing condition/helper recorded.	

CMAKE_C_COMPILER

cmake/toolchains/sdk.cmake:77

Field	Captured value
Value token 1	\${_c_compiler}
Context: no enclosing condition/helper recorded.	

CMAKE_SYSROOT

cmake/toolchains/sdk.cmake:78

Field	Captured value
Value token 1	\${_sysroot}
Context: no enclosing condition/helper recorded.	

CMAKE_FIND_ROOT_PATH

cmake/toolchains/sdk.cmake:79

Field	Captured value
Value token 1	\${CMAKE_SYSROOT}
Context: no enclosing condition/helper recorded.	

CMAKE_FIND_ROOT_PATH_MODE_PROGRAM

cmake/toolchains/sdk.cmake:80

Field	Captured value
Value token 1	NEVER
Context: no enclosing condition/helper recorded.	

CMAKE_FIND_ROOT_PATH_MODE_LIBRARY

cmake/toolchains/sdk.cmake:81

Field	Captured value
Value token 1	ONLY
Context: no enclosing condition/helper recorded.	

CMAKE_FIND_ROOT_PATH_MODE_INCLUDE

cmake/toolchains/sdk.cmake:82

Field	Captured value
Value token 1	ONLY
Context: no enclosing condition/helper recorded.	

CMAKE_FIND_ROOT_PATH_MODE_PACKAGE

cmake/toolchains/sdk.cmake:83

Field	Captured value
Value token 1	ONLY

Context: no enclosing condition/helper recorded.

CMAKE_FIND_USE_PACKAGE_REGISTRY

cmake/toolchains/sdk.cmake:84

Field	Captured value
Value token 1	OFF

Context: no enclosing condition/helper recorded.

CMAKE_FIND_USE_SYSTEM_PACKAGE_REGISTRY

cmake/toolchains/sdk.cmake:85

Field	Captured value
Value token 1	OFF

Context: no enclosing condition/helper recorded.

4. Configuration command reference

All 170 commands in the compiler/link/dependency command selection below are printed with full arguments and context. The other 1683 of 1853 captured commands remain in the complete HTML/JSON inventory. Target source/link declarations already have full detail in section 1; option defaults and CMAKE_* assignments are in sections 2-3.

Selection: [add_compile_definitions](#), [add_compile_options](#), [add_custom_command](#), [add_dependencies](#), [add_link_options](#), [add_subdirectory](#), [check_cxx_source_compiles](#), [cmake_minimum_required](#), [enable_language](#), [find_package](#), [find_program](#), [include](#), [include_directories](#), [link_directories](#), [link_libraries](#), [project](#), [set_property](#), [set_target_properties](#), [target_compile_definitions](#), [target_compile_features](#), [target_compile_options](#), [target_include_directories](#), [target_link_options](#)

[Open the complete compiler explorer](#) | [Captured JSON inventory](#)

CMakeLists.txt

cmake_minimum_required | [CMakeLists.txt:1](#)

Context: no enclosing condition/helper recorded.

```
cmake_minimum_required(VERSION 3.24)
```

project | [CMakeLists.txt:2](#)

Context: no enclosing condition/helper recorded.

```
project(Foundation VERSION 0.1.0 LANGUAGES CXX)
```

include | [CMakeLists.txt:7](#)

Context: no enclosing condition/helper recorded.

```
include(GNUInstallDirs)
```

include | [CMakeLists.txt:8](#)

Context: no enclosing condition/helper recorded.

```
include(CMakePackageConfigHelpers)
```

include | [CMakeLists.txt:9](#)

Context: no enclosing condition/helper recorded.

```
include(CTest)
```

include | [CMakeLists.txt:10](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/TestPrerequisites.cmake)
```

include | [CMakeLists.txt:11](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/BuildPolicy.cmake)
```

include | [CMakeLists.txt:15](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/StaticCxxRuntime.cmake)
```

include | [CMakeLists.txt:16](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/SdkBuildRuntime.cmake)
```

set_property | [CMakeLists.txt:32](#)

Context: no enclosing condition/helper recorded.

```
set_property(CACHE FOUNDATION_CORE_PROVIDER PROPERTY STRINGS cpp rust)
```

set_target_properties | [CMakeLists.txt:60](#)

Context: no enclosing condition/helper recorded.

```
set_target_properties(foundation_core PROPERTIES EXPORT_NAME core CXX_EXTENSIONS OFF)
```

target_compile_features | [CMakeLists.txt:61](#)

Context: no enclosing condition/helper recorded.

```
target_compile_features(foundation_core PUBLIC cxx_std_20)
```

target_include_directories | [CMakeLists.txt:62](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation_core PUBLIC
$<BUILD_INTERFACE:${CMAKE_CURRENT_SOURCE_DIR}/include>
$<BUILD_INTERFACE:${CMAKE_CURRENT_BINARY_DIR}/generated>
$<INSTALL_INTERFACE:${CMAKE_INSTALL_INCLUDEDIR}>)
```

target_compile_options | [CMakeLists.txt:70](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(EMSCRIPTEN)

```
target_compile_options(${target} PRIVATE -fexceptions)
```

target_link_options | [CMakeLists.txt:71](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(EMSCRIPTEN)

```
target_link_options(${target} PRIVATE -fexceptions -sASSERTIONS=1)
```

target_compile_options | [CMakeLists.txt:74](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(MSVC)

```
target_compile_options(${target} PRIVATE /w4 /permissive- /utf-8)
```

target_compile_options | [CMakeLists.txt:76](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(MSVC) -> else()

```
target_compile_options(${target} PRIVATE -Wall -Wextra -Wpedantic)
```

target_compile_options | [CMakeLists.txt:80](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_PORTABLE AND NOT MSVC AND NOT EMSCRIPTEN)
3. if(CMAKE_SYSTEM_PROCESSOR MATCHES ^(x86_64|AMD64|amd64)\$)

```
target_compile_options(${target} PRIVATE -march=x86-64 -mtune=generic)
```

target_compile_options | [CMakeLists.txt:82](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_PORTABLE AND NOT MSVC AND NOT EMSCRIPTEN)
3. if(CMAKE_SYSTEM_PROCESSOR MATCHES ^(x86_64|AMD64|amd64)\$ -> elseif(CMAKE_SYSTEM_PROCESSOR MATCHES ^(aarch64|arm64|ARM64)\$)


```
target_compile_options(${target} PRIVATE -march=armv8-a)
```

target_link_options | [CMakeLists.txt:89](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_PORTABLE AND NOT MSVC AND NOT EMSCRIPTEN)


```
target_link_options(${target} PRIVATE "-Wl,--exclude-libs,ALL" "-Wl,--disable-new-dtags")
```

set_target_properties | [CMakeLists.txt:90](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_PORTABLE AND NOT MSVC AND NOT EMSCRIPTEN)


```
set_target_properties(${target} PROPERTIES INSTALL_RPATH "$ORIGIN/../lib/runtime"
BUILD_RPATH_USE_ORIGIN ON)
```

target_compile_options | [CMakeLists.txt:96](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_SANITIZERS)


```
target_compile_options(${target} PRIVATE -fsanitize=address,undefined -fno-omit-frame-pointer)
```

target_link_options | [CMakeLists.txt:97](#)

Enclosing context (unevaluated):

1. function(foundation_options target)
2. if(FOUNDATION_SANITIZERS)


```
target_link_options(${target} PRIVATE -fsanitize=address,undefined)
```

target_link_options | [CMakeLists.txt:102](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_SANITIZERS)


```
target_link_options(foundation_core INTERFACE -fsanitize=address,undefined)
```

target_include_directories | [CMakeLists.txt:107](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation-cli PRIVATE "${CMAKE_CURRENT_BINARY_DIR}/generated")
```

target_link_options | [CMakeLists.txt:110](#)

Enclosing context (unevaluated):

1. if(EMSCRIPTEN)


```
target_link_options(foundation-cli PRIVATE -sENVIRONMENT=node
"-sINCOMING_MODULE_JS_API=['print','printErr']"
--pre-js "${CMAKE_CURRENT_SOURCE_DIR}/src/platform/node_console.js")
```

set_property | [CMakeLists.txt:113](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    set_property(TARGET foundation-cli APPEND PROPERTY LINK_DEPENDS
        "${CMAKE_CURRENT_SOURCE_DIR}/src/platform/node_console.js")
```

find_package | [CMakeLists.txt:117](#)

Context: no enclosing condition/helper recorded.

```
find_package(Python3 3.9 REQUIRED COMPONENTS Interpreter)
```

include | [CMakeLists.txt:118](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/RustComponent.cmake)
```

include | [CMakeLists.txt:120](#)

Context: no enclosing condition/helper recorded.

```
include(cmake/ImportWasm.cmake)
```

target_include_directories | [CMakeLists.txt:165](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(WIN32 AND MSVC AND NOT CMAKE_CROSSCOMPILING)
    target_include_directories(foundation_windows_arguments_test PRIVATE
        "${CMAKE_CURRENT_SOURCE_DIR}/src")
```

target_compile_features | [CMakeLists.txt:166](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(WIN32 AND MSVC AND NOT CMAKE_CROSSCOMPILING)
    target_compile_features(foundation_windows_arguments_test PRIVATE cxx_std_20)
```

target_include_directories | [CMakeLists.txt:178](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
    target_include_directories(foundation_rust_component_test PRIVATE
        "${CMAKE_CURRENT_SOURCE_DIR}/src")
```

target_compile_definitions | [CMakeLists.txt:180](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(FOUNDATION_CORE_PROVIDER STREQUAL rust)
    target_compile_definitions(foundation_rust_component_test PRIVATE FOUNDATION_EXPECT_RUST=1)
```

target_include_directories | [CMakeLists.txt:191](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
    target_include_directories(foundation_text_status_test PRIVATE
        "${CMAKE_CURRENT_SOURCE_DIR}/include" "${CMAKE_CURRENT_SOURCE_DIR}/src")
```

target_compile_features | [CMakeLists.txt:192](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
    target_compile_features(foundation_text_status_test PRIVATE cxx_std_20)
```

find_program | [CMakeLists.txt:231](#)

Enclosing context (unevaluated):

```
1. if(BUILD_TESTING)
2. if(FOUNDATION_DISTRIBUTION_TESTS)
3. foreach(_tool dpkg-deb gpg gpgv gpgconf gpg-agent gpg-connect-agent git cc)
    find_program(_required_${_tool} NAMES ${_tool} REQUIRED)
```

add_subdirectory | [CMakeLists.txt:261](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_BUILD_GUI)
    add_subdirectory(gui)
```

set_property | [CMakeLists.txt:330](#)

Context: no enclosing condition/helper recorded.

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS ${_source_paths})
```

include | [CMakeLists.txt:402](#)

Context: no enclosing condition/helper recorded.

```
include(CPack)
```

cmake/BuildPolicy.cmake**include** | [cmake/BuildPolicy.cmake:2](#)

Context: no enclosing condition/helper recorded.

```
include("${CMAKE_CURRENT_LIST_DIR}/DependencyPaths.cmake")
```

find_package | [cmake/BuildPolicy.cmake:8](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_DEPENDENCY_PREFIX)
    find_package(Python3 3.9 REQUIRED COMPONENTS Interpreter)
```

include_directories | [cmake/BuildPolicy.cmake:19](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_DEPENDENCY_PREFIX)
    include_directories(SYSTEM "${FOUNDATION_DEPENDENCY_PREFIX}/usr/include")
```

include_directories | [cmake/BuildPolicy.cmake:21](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_DEPENDENCY_PREFIX)
2. if(CMAKE_LIBRARY_ARCHITECTURE AND IS_DIRECTORY
${FOUNDATION_DEPENDENCY_PREFIX}/usr/include/${CMAKE_LIBRARY_ARCHITECTURE})
    include_directories(SYSTEM
"${FOUNDATION_DEPENDENCY_PREFIX}/usr/include/${CMAKE_LIBRARY_ARCHITECTURE}")
```

find_program | [cmake/BuildPolicy.cmake:30](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_COMPILER_CACHE AND CMAKE_GENERATOR MATCHES Ninja|Makefiles)
 find_program(FOUNDATION_CCACHE_PROGRAM NAMES ccache NO_CMAKE_FIND_ROOT_PATH)

set_property | [cmake/BuildPolicy.cmake:45](#)

Enclosing context (unevaluated):

1. function(foundation_notice_file path)
 set_property(GLOBAL APPEND PROPERTY FOUNDATION_NOTICE_FILES "\${path}")

set_property | [cmake/BuildPolicy.cmake:46](#)

Enclosing context (unevaluated):

1. function(foundation_notice_file path)
 set_property(GLOBAL APPEND PROPERTY FOUNDATION_NOTICE_HASHES "\${identity}")

set_property | [cmake/BuildPolicy.cmake:75](#)

Enclosing context (unevaluated):

1. function(foundation_generated_application_archive target producer)
2. foreach(configuration IN LISTS configurations)
 set_property(GLOBAL APPEND PROPERTY FOUNDATION_GENERATED_ARCHIVES "\${archive}")

set_property | [cmake/BuildPolicy.cmake:77](#)

Enclosing context (unevaluated):

1. function(foundation_generated_application_archive target producer)
 set_property(TARGET "\${target}" PROPERTY FOUNDATION_GENERATED_APPLICATION_ARCHIVE TRUE)

set_property | [cmake/BuildPolicy.cmake:78](#)

Enclosing context (unevaluated):

1. function(foundation_generated_application_archive target producer)
 set_property(GLOBAL APPEND PROPERTY FOUNDATION_GENERATED_TARGETS "\${target}")

set_property | [cmake/BuildPolicy.cmake:79](#)

Enclosing context (unevaluated):

1. function(foundation_generated_application_archive target producer)
 set_property(GLOBAL APPEND PROPERTY FOUNDATION_GENERATED_PRODUCERS "\${producer}")

set_property | [cmake/BuildPolicy.cmake:90](#)

Enclosing context (unevaluated):

1. function(foundation_register_build_directory)
 set_property(DIRECTORY PROPERTY FOUNDATION_POLICY_REGISTERED TRUE)

set_property | [cmake/BuildPolicy.cmake:105](#)

Enclosing context (unevaluated):

1. function(_foundation_capture_directory_policy)
 set_property(DIRECTORY PROPERTY FOUNDATION_POLICY_CAPTURED TRUE)

set_property | [cmake/BuildPolicy.cmake:122](#)

Enclosing context (unevaluated):

1. function(_foundation_capture_directory_policy)

```
set_property(GLOBAL APPEND PROPERTY FOUNDATION_NOTICE_FRAGMENTS "${notice_fragment}")
```

set_property | [cmake/BuildPolicy.cmake:221](#)

Enclosing context (unevaluated):

1. function(_foundation_capture_directory_policy)
2. if(FOUNDATION_SDK_ROOT AND CMAKE_SYSTEM_NAME STREQUAL Linux)

```
set_property(GLOBAL APPEND PROPERTY FOUNDATION_DEPENDENCY_FRAGMENTS "${check_file}")
```

add_dependencies | [cmake/BuildPolicy.cmake:298](#)

Enclosing context (unevaluated):

1. function(foundation_finalize_build_policy)
2. if(FOUNDATION_DEPENDENCY_PREFIX)
3. foreach(target IN LISTS targets)
4. if(NOT imported AND NOT type STREQUAL INTERFACE_LIBRARY AND NOT type STREQUAL UTILITY)

```
add_dependencies("${target}" foundation-prefix-check)
```

add_dependencies | [cmake/BuildPolicy.cmake:318](#)

Enclosing context (unevaluated):

1. function(foundation_finalize_build_policy)
2. if(FOUNDATION_SDK_ROOT AND CMAKE_SYSTEM_NAME STREQUAL Linux)
3. if(generated_producers)

```
add_dependencies(foundation-dependency-check ${generated_producers})
```

add_dependencies | [cmake/BuildPolicy.cmake:324](#)

Enclosing context (unevaluated):

1. function(foundation_finalize_build_policy)
2. if(FOUNDATION_SDK_ROOT AND CMAKE_SYSTEM_NAME STREQUAL Linux)
3. foreach(target IN LISTS targets)
4. if(NOT imported AND NOT type STREQUAL INTERFACE_LIBRARY AND NOT type STREQUAL UTILITY)

```
add_dependencies("${target}" foundation-dependency-check)
```

cmake/FoundationConfig.cmake.in

set_target_properties | [cmake/FoundationConfig.cmake.in:6](#)

Enclosing context (unevaluated):

1. if(Foundation_CORE_PROVIDER STREQUAL rust AND NOT TARGET foundation::rust_component)

```
set_target_properties(foundation::rust_component PROPERTIES
IMPORTED_LOCATION "${_foundation_rust_archive}"
INTERFACE_LINK_LIBRARIES "@FOUNDATION_RUST_NATIVE_LIBRARIES@"
INTERFACE_LINK_OPTIONS "@FOUNDATION_RUST_LINK_OPTIONS@")
```

include | [cmake/FoundationConfig.cmake.in:12](#)

Context: no enclosing condition/helper recorded.

```
include("${CMAKE_CURRENT_LIST_DIR}/FoundationTargets.cmake")
```

include | [cmake/FoundationConfig.cmake.in:13](#)

Context: no enclosing condition/helper recorded.

```
include("${CMAKE_CURRENT_LIST_DIR}/StaticCxxRuntime.cmake")
```

target_compile_options | [cmake/FoundationConfig.cmake.in:20](#)

Enclosing context (unevaluated):

1. function(foundation_apply_runtime target)
2. if(EMSCRIPTEN)

```
target_compile_options(${target} PRIVATE -fexceptions)
```

target_link_options | [cmake/FoundationConfig.cmake.in:21](#)

Enclosing context (unevaluated):

1. function(foundation_apply_runtime target)
2. if(EMSCRIPTEN)

```
target_link_options(${target} PRIVATE -fexceptions -sASSERTIONS=1)
```

set_property | [cmake/FoundationConfig.cmake.in:23](#)

Enclosing context (unevaluated):

1. function(foundation_apply_runtime target)
2. if(EMSCRIPTEN) -> elseif(MSVC AND Foundation_MSVC_RUNTIME_LIBRARY)

```
set_property(TARGET ${target} PROPERTY MSVC_RUNTIME_LIBRARY "${Foundation_MSVC_RUNTIME_LIBRARY}")
```

cmake/ImportWasm.cmake

add_dependencies | [cmake/ImportWasm.cmake:28](#)

Enclosing context (unevaluated):

1. if(TARGET foundation_core)

```
add_dependencies(foundation_core foundation-wasm-import)
```

set_property | [cmake/ImportWasm.cmake:32](#)

Context: no enclosing condition/helper recorded.

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
"${FOUNDATION_WASM_PACKAGE}/web-manifest.json" "${FOUNDATION_WASM_PACKAGE}/manifest.sha256"
"${FOUNDATION_WASM_PACKAGE}/software-foundation-wasm.html")
```

cmake/RustComponent.cmake

set_property | [cmake/RustComponent.cmake:58](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. if(FOUNDATION_RUST_SDK_ROOT)

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
"${FOUNDATION_RUST_SDK_ROOT}/rust-sdk.json")
```

find_program | [cmake/RustComponent.cmake:63](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. if(FOUNDATION_RUST_SDK_ROOT) -> else()

```
find_program(FOUNDATION_RUST_RUSTC NAMES rustc NO_CMAKE_FIND_ROOT_PATH)
```

find_program | [cmake/RustComponent.cmake:64](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. if(FOUNDATION_RUST_SDK_ROOT) -> else()


```
find_program(FOUNDATION_RUST_CARGO NAMES cargo NO_CMAKE_FIND_ROOT_PATH)
```

set_property | [cmake/RustComponent.cmake:104](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. foreach(configuration IN LISTS configurations)


```
set_property(TARGET foundation::rust_component APPEND PROPERTY IMPORTED_CONFIGURATIONS
"${configuration}")
```

set_property | [cmake/RustComponent.cmake:105](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. foreach(configuration IN LISTS configurations)


```
set_property(TARGET foundation::rust_component PROPERTY "IMPORTED_LOCATION_${upper}" "${archive}")
```

set_property | [cmake/RustComponent.cmake:145](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)


```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS ${all_inputs} ${rust_inputs}
"${CMAKE_SOURCE_DIR}/tools/rust_build.py" "${CMAKE_SOURCE_DIR}/tools/rust_sdk.py")
```

add_dependencies | [cmake/RustComponent.cmake:155](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)


```
add_dependencies(foundation::rust_component foundation-rust)
```

set_target_properties | [cmake/RustComponent.cmake:162](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)


```
set_target_properties(foundation::rust_component PROPERTIES
FOUNDATION_GENERATED_RECEIPTS "${receipts}" FOUNDATION_GENERATED_INPUTS "${all_inputs}"
FOUNDATION_GENERATED_VERIFIER "${Python3_EXECUTABLE}; -B;${CMAKE_SOURCE_DIR}/tools/rust_build.p
y;verify;--config;${selected_config}"
INTERFACE_LINK_LIBRARIES "${native_libraries}")
```

set_property | [cmake/RustComponent.cmake:169](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)
2. if(CMAKE_SYSTEM_NAME STREQUAL Linux)


```
set_property(TARGET foundation::rust_component PROPERTY INTERFACE_LINK_OPTIONS "${link_options}")
```

target_compile_definitions | [cmake/RustComponent.cmake:172](#)

Enclosing context (unevaluated):

1. function(foundation_rust_component core)


```
target_compile_definitions(${core} PRIVATE FOUNDATION_USE_RUST=1)
```

cmake/SdkBuildRuntime.cmake

set_property | cmake/SdkBuildRuntime.cmake:22

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)
2. if(NOT FOUNDATION_PORTABLE)
3. if(NOT checked)

```
set_property(GLOBAL PROPERTY FOUNDATION_SDK_STATIC_RUNTIME_CHECKED TRUE)
```

set_property | cmake/SdkBuildRuntime.cmake:37

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)

```
set_property(TARGET "${target}" PROPERTY FOUNDATION_SDK_RUNTIME TRUE)
```

target_link_options | cmake/SdkBuildRuntime.cmake:43

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)

```
target_link_options(${target} PRIVATE "LINKER:--disable-new-dtags")
```

set_target_properties | cmake/SdkBuildRuntime.cmake:44

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)

```
set_target_properties(${target} PROPERTIES SKIP_BUILD_RPATH TRUE
BUILD_WITH_INSTALL_RPATH TRUE)
```

set_property | cmake/SdkBuildRuntime.cmake:46

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)

```
set_property(GLOBAL APPEND PROPERTY FOUNDATION_SDK_ALL_RUNTIME_TARGETS "${target}")
```

set_property | cmake/SdkBuildRuntime.cmake:47

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)

```
set_property(DIRECTORY APPEND PROPERTY FOUNDATION_SDK_RUNTIME_TARGETS "${target}")
```

set_property | cmake/SdkBuildRuntime.cmake:50

Enclosing context (unevaluated):

1. function(foundation_sdk_runtime target)
2. if(NOT scheduled)

```
set_property(DIRECTORY PROPERTY FOUNDATION_SDK_RUNTIME_SCHEDULED TRUE)
```

set_target_properties | cmake/SdkBuildRuntime.cmake:95

Enclosing context (unevaluated):

1. function(_foundation_finalize_sdk_runtime)
2. foreach(target IN LISTS targets)

```
set_target_properties(${target} PROPERTIES FOUNDATION_SDK_INSTALL_RPATH "${install_rpath}"
INSTALL_RPATH "$ORIGIN/.sdk-runtime/${target}" INSTALL_RPATH_USE_LINK_PATH FALSE)
```

add_dependencies | [cmake/SdkBuildRuntime.cmake:105](#)

Enclosing context (unevaluated):

1. function(_ foundation_finalize_sdk_runtime)
2. foreach(target IN LISTS targets)


```
add_dependencies("${target}" "foundation-sdk-runtime-${target}")
```

add_custom_command | [cmake/SdkBuildRuntime.cmake:106](#)

Enclosing context (unevaluated):

1. function(_ foundation_finalize_sdk_runtime)
2. foreach(target IN LISTS targets)


```
add_custom_command(TARGET "${target}" POST_BUILD
COMMAND "${Python3_EXECUTABLE}" -B "${helper}"
--sdk "${FOUNDATION_SDK_ROOT}" --manifest-sha256 "${manifest}"
--executable "$<TARGET_FILE:${target}>" --target "${target}"
--processor "${CMAKE_SYSTEM_PROCESSOR}"
COMMENT "Auditing private SDK development runtime for ${target}" VERBATIM)
```

set_property | [cmake/SdkBuildRuntime.cmake:112](#)

Enclosing context (unevaluated):

1. function(_ foundation_finalize_sdk_runtime)
2. foreach(target IN LISTS targets)


```
set_property(TARGET "${target}" APPEND PROPERTY LINK_DEPENDS
"${helper}" "${CMAKE_CURRENT_FUNCTION_LIST_DIR}/../tools/stage_runtime.py"
"${CMAKE_CURRENT_FUNCTION_LIST_DIR}/../tools/verify_abi.py"
"${CMAKE_CURRENT_FUNCTION_LIST_FILE}")
```

cmake/StaticCxxRuntime.cmake**include** | [cmake/StaticCxxRuntime.cmake:11](#)

Enclosing context (unevaluated):

1. function(foundation_require_static_gnu_runtime)


```
include(CheckCXXSourceCompiles)
```

check_cxx_source_compiles | [cmake/StaticCxxRuntime.cmake:15](#)

Enclosing context (unevaluated):

1. function(foundation_require_static_gnu_runtime)


```
check_cxx_source_compiles("#include <cstdint>
#ifndef __GLIBCXX__
#error The selected standard library is not libstdc++
#endif
int main() { return 0; }" foundation_selected_gnu_runtime)
```

target_link_options | [cmake/StaticCxxRuntime.cmake:28](#)

Enclosing context (unevaluated):

1. function(foundation_link_static_gnu_runtime target visibility)


```
target_link_options(${target} ${visibility} -static-libstdc++ -static-libgcc
"LINKER:--exclude-libs,libstdc++.a:libgcc.a:libgcc_eh.a")
```

cmake/TestPrerequisites.cmake

set_property | [cmake/TestPrerequisites.cmake:15](#)

Enclosing context (unevaluated):

1. function(foundation_test_prerequisites name)

```
set_property(GLOBAL APPEND PROPERTY FOUNDATION_REGISTERED_TESTS "${name}")
```

set_property | [cmake/TestPrerequisites.cmake:16](#)

Enclosing context (unevaluated):

1. function(foundation_test_prerequisites name)

```
set_property(GLOBAL PROPERTY "FOUNDATION_TEST_${key}_DIRECTORY" "${CMAKE_CURRENT_SOURCE_DIR}")
```

set_property | [cmake/TestPrerequisites.cmake:17](#)

Enclosing context (unevaluated):

1. function(foundation_test_prerequisites name)

```
set_property(GLOBAL PROPERTY "FOUNDATION_TEST_${key}_TARGETS" "${ARGN}")
```

set_property | [cmake/TestPrerequisites.cmake:19](#)

Enclosing context (unevaluated):

1. function(foundation_test_prerequisites name)

```
set_property(DIRECTORY APPEND PROPERTY FOUNDATION_DIRECTORY_TESTS "${name}")
```

set_property | [cmake/TestPrerequisites.cmake:33](#)

Enclosing context (unevaluated):

1. function(_foundation_capture_test_labels)

2. foreach(name IN LISTS names)

```
set_property(GLOBAL PROPERTY "FOUNDATION_TEST_${key}_LABELS" "${labels}")
```

add_dependencies | [cmake/TestPrerequisites.cmake:85](#)

Enclosing context (unevaluated):

1. function(foundation_finalize_test_prerequisites)

2. if(TARGET foundation-gui-tests)

```
add_dependencies(foundation-gui-tests foundation-tests-gui)
```

cmake/toolchains/sdk.cmake

find_program | [cmake/toolchains/sdk.cmake:9](#)

Enclosing context (unevaluated):

1. if(NOT _nested)

```
find_program(_sdk_python NAMES python3 python REQUIRED NO_CMAKE_FIND_ROOT_PATH)
```

include | [cmake/toolchains/sdk.cmake:39](#)

Enclosing context (unevaluated):

1. if(_system STREQUAL Emscripten)

```
include("${_wasm_toolchain}")
```

include | [cmake/toolchains/sdk.cmake:43](#)

Enclosing context (unevaluated):

```
1. if(_system STREQUAL Emscripten)
    include("${CMAKE_USER_MAKE_RULES_OVERRIDE}")
```

examples/consumer/CMakeLists.txt

cmake_minimum_required | [examples/consumer/CMakeLists.txt:1](#)

Context: no enclosing condition/helper recorded.

```
cmake_minimum_required(VERSION 3.24)
```

project | [examples/consumer/CMakeLists.txt:2](#)

Context: no enclosing condition/helper recorded.

```
project(FoundationConsumer LANGUAGES CXX)
```

find_package | [examples/consumer/CMakeLists.txt:3](#)

Context: no enclosing condition/helper recorded.

```
find_package(Foundation 0.1 CONFIG REQUIRED)
```

target_compile_definitions | [examples/consumer/CMakeLists.txt:11](#)

Enclosing context (unevaluated):

```
1. if(DEFINED Foundation_CORE_PROVIDER)
2. if(Foundation_CORE_PROVIDER STREQUAL rust)
    target_compile_definitions(consumer PRIVATE FOUNDATION_EXPECT_TEXT_PROVIDER=1)
```

target_compile_definitions | [examples/consumer/CMakeLists.txt:13](#)

Enclosing context (unevaluated):

```
1. if(DEFINED Foundation_CORE_PROVIDER)
2. if(Foundation_CORE_PROVIDER STREQUAL rust) -> elseif(Foundation_CORE_PROVIDER STREQUAL cpp)
    target_compile_definitions(consumer PRIVATE FOUNDATION_EXPECT_TEXT_PROVIDER=0)
```

gui/CMakeLists.txt

set_property | [gui/CMakeLists.txt:20](#)

Enclosing context (unevaluated):

```
1. if(FOUNDATION_GUI_INPUT_GROUP)
    set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
        "${FOUNDATION_GUI_INPUT_GROUP}/manifest.json" "${FOUNDATION_GUI_INPUT_GROUP}/SHA256SUMS"
        "${FOUNDATION_GUI_INPUT_GROUP}/gui-inputs.tar.gz"
        "${CMAKE_CURRENT_SOURCE_DIR}/source_group.py"
        "${foundation_gui_project_root}/tools/dependency_archive.py")
```

set_property | [gui/CMakeLists.txt:69](#)

Context: no enclosing condition/helper recorded.

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS "${lock_path}" ${verified_files})
```

find_package | [gui/CMakeLists.txt:71](#)

Enclosing context (unevaluated):

```
1. if(NOT EMSCRIPTEN)
    find_package(Threads REQUIRED)
```

target_include_directories | [gui/CMakeLists.txt:75](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation_gui_boundary INTERFACE
"${CMAKE_CURRENT_BINARY_DIR}/include")
```

target_include_directories | [gui/CMakeLists.txt:76](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation_gui_boundary SYSTEM INTERFACE
"${FOUNDATION_GUI_SOURCE}/include")
```

target_compile_features | [gui/CMakeLists.txt:77](#)

Context: no enclosing condition/helper recorded.

```
target_compile_features(foundation_gui_boundary INTERFACE cxx_std_20)
```

target_compile_definitions | [gui/CMakeLists.txt:80](#)

Enclosing context (unevaluated):

1. if(WIN32)

```
target_compile_definitions(foundation_gui_boundary INTERFACE NOMINMAX)
```

set_property | [gui/CMakeLists.txt:92](#)

Context: no enclosing condition/helper recorded.

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
${foundation_gui_transitive_sources})
```

add_custom_command | [gui/CMakeLists.txt:93](#)

Context: no enclosing condition/helper recorded.

```
add_custom_command(OUTPUT "${CMAKE_CURRENT_BINARY_DIR}/boundary-checked"
COMMAND "${Python3_EXECUTABLE}" -B "${CMAKE_CURRENT_SOURCE_DIR}/check_boundary.py"
COMMAND "${CMAKE_COMMAND}" -E touch "${CMAKE_CURRENT_BINARY_DIR}/boundary-checked"
DEPENDS ${foundation_gui_owned_sources} ${foundation_gui_transitive_sources}
"${CMAKE_CURRENT_SOURCE_DIR}/check_boundary.py" VERBATIM)
```

add_dependencies | [gui/CMakeLists.txt:105](#)

Context: no enclosing condition/helper recorded.

```
add_dependencies(foundation_gui_application foundation_gui_contract)
```

target_include_directories | [gui/CMakeLists.txt:107](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation_gui_application PUBLIC "${CMAKE_CURRENT_SOURCE_DIR}")
```

set_target_properties | [gui/CMakeLists.txt:108](#)

Context: no enclosing condition/helper recorded.

```
set_target_properties(foundation_gui_application PROPERTIES CXX_EXTENSIONS OFF)
```

set_target_properties | [gui/CMakeLists.txt:120](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)

2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)

```
set_target_properties(${target} PROPERTIES WIN32_EXECUTABLE TRUE)
```

target_compile_definitions | [gui/CMakeLists.txt:121](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)
 target_compile_definitions(\${target} PRIVATE main=foundation_gui_main SDL_MAIN_HANDLED)

target_compile_definitions | [gui/CMakeLists.txt:123](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)
3. if(target STREQUAL foundation-gui-sdl)
 target_compile_definitions(\${target} PRIVATE FOUNDATION_GUI_SDL_ENTRY)

target_link_options | [gui/CMakeLists.txt:127](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)
3. if(MSVC)
 target_link_options(\${target} PRIVATE /MANIFEST:EMBED
 "/MANIFESTINPUT:\${CMAKE_CURRENT_SOURCE_DIR}/hosts/windows_gui.manifest")

enable_language | [gui/CMakeLists.txt:130](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)
3. if(MSVC) -> else()
 enable_language(RC)

target_include_directories | [gui/CMakeLists.txt:131](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(WIN32 AND target MATCHES ^foundation-gui-(fltk|rev|sdl)\$)
3. if(MSVC) -> else()
 target_include_directories(\${target} PRIVATE "\${CMAKE_CURRENT_SOURCE_DIR}/hosts")

target_include_directories | [gui/CMakeLists.txt:136](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
 target_include_directories(\${target} PRIVATE "\${CMAKE_CURRENT_BINARY_DIR}")

add_dependencies | [gui/CMakeLists.txt:139](#)

Enclosing context (unevaluated):

1. function(foundation_gui_executable target source)
2. if(target MATCHES ^foundation-gui-)
 add_dependencies(foundation-gui-hosts \${target})

set_property | [gui/CMakeLists.txt:151](#)

Enclosing context (unevaluated):

```
1. function(foundation_gui_patch source patch output)
    set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
        "${CMAKE_CURRENT_SOURCE_DIR}/patches/${patch}"
        "${CMAKE_CURRENT_SOURCE_DIR}/patches/apply.py")
```

set_property | [gui/CMakeLists.txt:162](#)

Enclosing context (unevaluated):

```
1. function(foundation_gui_repatch source patch output)
    set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
        "${CMAKE_CURRENT_SOURCE_DIR}/patches/${patch}")
```

set_target_properties | [gui/CMakeLists.txt:166](#)

Enclosing context (unevaluated):

```
1. function(foundation_gui_check name target labels)
    set_target_properties(${target} PROPERTIES EXCLUDE_FROM_ALL TRUE)
```

find_program | [gui/CMakeLists.txt:186](#)

Context: no enclosing condition/helper recorded.

```
find_program(FOUNDATION_NODE_EXECUTABLE NAMES node nodejs)
```

set_property | [gui/CMakeLists.txt:280](#)

Context: no enclosing condition/helper recorded.

```
set_property(DIRECTORY APPEND PROPERTY CMAKE_CONFIGURE_DEPENDS
    "${CMAKE_SOURCE_DIR}/tools/browser_bundle.py")
```

set_target_properties | [gui/CMakeLists.txt:307](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    set_target_properties(foundation-gui-web-wasm PROPERTIES OUTPUT_NAME gui_web_wasm SUFFIX .js)
```

target_compile_options | [gui/CMakeLists.txt:308](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    target_compile_options(foundation_gui_application PRIVATE -fexceptions)
```

target_compile_options | [gui/CMakeLists.txt:309](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    target_compile_options(foundation-gui-web-wasm PRIVATE -fexceptions)
```

target_link_options | [gui/CMakeLists.txt:310](#)

Enclosing context (unevaluated):

```
1. if(EMSCRIPTEN)
    target_link_options(foundation-gui-web-wasm PRIVATE --no-entry -fexceptions
        -sMODULARIZE=1 -sEXPORT_ES6=1 -sALLOW_MEMORY_GROWTH=1
        "-sEXPORTED_FUNCTIONS=['_gui_web_create','_gui_web_receive','_gui_web_destroy']"
        "-sEXPORTED_RUNTIME_METHODS=['ccall']" -sENVIRONMENT=web,worker,node)
```

target_link_options | [gui/CMakeLists.txt:314](#)

Enclosing context (unevaluated):

1. if(EMSCRIPTEN)

```
target_link_options(foundation-gui-web-wasm PRIVATE
  "-sINCOMING_MODULE_JS_API=['wasmBinary','locateFile','print','printErr','onRuntimeInitialized']")
```

add_custom_command | [gui/CMakeLists.txt:333](#)

Enclosing context (unevaluated):

1. if(EMSCRIPTEN)

```
add_custom_command(OUTPUT "${wasm_html}" "${wasm_manifest}" "${wasm_checksums}"
  COMMAND "${Python3_EXECUTABLE}" -B "${CMAKE_SOURCE_DIR}/tools/package_wasm.py"
  --assets "${CMAKE_CURRENT_BINARY_DIR}/web"
  --module-dir "$<TARGET_FILE_DIR:foundation-gui-web-wasm>"
  --output "${wasm_package}" --source-root "${CMAKE_SOURCE_DIR}" ${wasm_notice_args}
  DEPENDS foundation-gui-web-wasm "${CMAKE_CURRENT_BINARY_DIR}/wasm-package-source.txt"
  "${CMAKE_SOURCE_DIR}/tools/source_identity.py"
  "${CMAKE_SOURCE_DIR}/tools/package_wasm.py"
  "${CMAKE_SOURCE_DIR}/tools/package_notices.py"
  "${CMAKE_SOURCE_DIR}/tools/sdk_manifest.py"
  "${CMAKE_SOURCE_DIR}/tools/browser_bundle.py" ${foundation_web_assets}
  ${wasm_notices} VERBATIM)
```

find_package | [gui/CMakeLists.txt:372](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

```
find_package(FLTK REQUIRED)
```

target_include_directories | [gui/CMakeLists.txt:375](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

```
target_include_directories(foundation_gui_fltk SYSTEM INTERFACE
  "${CMAKE_CURRENT_BINARY_DIR}/fltk" ${FLTK_INCLUDE_DIR})
```

find_package | [gui/CMakeLists.txt:381](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_FLTK)

2. if(FOUNDATION_SDK_ROOT AND CMAKE_SYSTEM_NAME STREQUAL Linux)

```
find_package(X11 REQUIRED COMPONENTS Xft Xrender)
```

find_package | [gui/CMakeLists.txt:388](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_SDL)

```
find_package(SDL2 CONFIG REQUIRED)
```

include | [gui/CMakeLists.txt:416](#)

Enclosing context (unevaluated):

1. if(FOUNDATION_GUI_REV)

```
include("${CMAKE_CURRENT_BINARY_DIR}/Rev.cmake")
```

target_include_directories | [gui/CMakeLists.txt:424](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_SDL)
 target_include_directories(foundation_gui_sdl_touch_test PRIVATE "\${CMAKE_CURRENT_BINARY_DIR}")

target_include_directories | [gui/CMakeLists.txt:454](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(example touch embedding)
 target_include_directories(foundation_gui_\${example}_test PRIVATE "\${CMAKE_CURRENT_SOURCE_DIR}")

target_include_directories | [gui/CMakeLists.txt:503](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(test_name contract bitmap adapter layout runtime presentation extension interaction framebuffer terminal web)
 target_include_directories(\${test_target} PRIVATE "\${FOUNDATION_GUI_SOURCE}/examples"
 "\${FOUNDATION_GUI_SOURCE}/tests")

target_compile_options | [gui/CMakeLists.txt:505](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(test_name contract bitmap adapter layout runtime presentation extension interaction framebuffer terminal web)
3. if(MSVC)
 target_compile_options(\${test_target} PRIVATE /UNDEBUG /utf-8)

target_compile_options | [gui/CMakeLists.txt:507](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. foreach(test_name contract bitmap adapter layout runtime presentation extension interaction framebuffer terminal web)
3. if(MSVC) -> else()
 target_compile_options(\${test_target} PRIVATE -UNDEBUG)

target_include_directories | [gui/CMakeLists.txt:558](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_HOST_TESTS)
3. foreach(backend fltk rev)
4. if(FOUNDATION_GUI_\${upper})
 target_include_directories(\${supplier_target} PRIVATE "\${FOUNDATION_GUI_SOURCE}/examples")

target_compile_options | [gui/CMakeLists.txt:583](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_HOST_TESTS)
3. if(FOUNDATION_GUI_REV)
4. if(MSVC)

```
target_compile_options(foundation_gui_supplier_clipboard PRIVATE /UNDEBUG)
```

target_compile_options | [gui/CMakeLists.txt:585](#)

Enclosing context (unevaluated):

1. if(BUILD_TESTING)
2. if(FOUNDATION_GUI_HOST_TESTS)
3. if(FOUNDATION_GUI_REV)
4. if(MSVC) -> else()

```
target_compile_options(foundation_gui_supplier_clipboard PRIVATE -UNDEBUG)
```

tests/cxx_runtime/CMakeLists.txt

cmake_minimum_required | [tests/cxx_runtime/CMakeLists.txt:1](#)

Context: no enclosing condition/helper recorded.

```
cmake_minimum_required(VERSION 3.24)
```

project | [tests/cxx_runtime/CMakeLists.txt:2](#)

Context: no enclosing condition/helper recorded.

```
project(FoundationRuntimeFixture LANGUAGES CXX)
```

include | [tests/cxx_runtime/CMakeLists.txt:6](#)

Context: no enclosing condition/helper recorded.

```
include("${FOUNDATION_SOURCE}/cmake/StaticCxxRuntime.cmake")
```

target_compile_features | [tests/cxx_runtime/CMakeLists.txt:9](#)

Context: no enclosing condition/helper recorded.

```
target_compile_features(host_plugin PRIVATE cxx_std_17)
```

include | [tests/cxx_runtime/CMakeLists.txt:12](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package)

```
include(CMakePackageConfigHelpers)
```

target_compile_features | [tests/cxx_runtime/CMakeLists.txt:15](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package)

```
target_compile_features(foundation_core PUBLIC cxx_std_20)
```

target_include_directories | [tests/cxx_runtime/CMakeLists.txt:16](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package)

```
target_include_directories(foundation_core PUBLIC
"$<BUILD_INTERFACE:${FOUNDATION_SOURCE}/include>" "$<INSTALL_INTERFACE:include>")
```

set_target_properties | [tests/cxx_runtime/CMakeLists.txt:18](#)

Enclosing context (unevaluated):

1. if(MODE STREQUAL package)

```
set_target_properties(foundation_core PROPERTIES EXPORT_NAME core)
```

target_compile_features | [tests/cxx_runtime/CMakeLists.txt:30](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
    target_compile_features(runtime_probe PRIVATE cxx_std_17)
```

target_link_options | [tests/cxx_runtime/CMakeLists.txt:34](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
    target_link_options(runtime_probe PRIVATE "LINKER:--export-dynamic" "LINKER:--disable-new-dtags")
```

set_target_properties | [tests/cxx_runtime/CMakeLists.txt:35](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
    set_target_properties(runtime_probe PROPERTIES INSTALL_RPATH "$ORIGIN/./lib")
```

target_link_options | [tests/cxx_runtime/CMakeLists.txt:40](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
2. if(MODE STREQUAL protected)
    target_link_options(runtime_probe PRIVATE "LINKER:--exclude-libs,ALL")
```

target_link_options | [tests/cxx_runtime/CMakeLists.txt:42](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
2. if(MODE STREQUAL protected) -> elseif(MODE STREQUAL unprotected)
    target_link_options(runtime_probe PRIVATE -static-libstdc++ -static-libgcc)
```

find_package | [tests/cxx_runtime/CMakeLists.txt:44](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
2. if(MODE STREQUAL protected) -> elseif(MODE STREQUAL installed)
    find_package(Foundation REQUIRED CONFIG)
```

target_compile_definitions | [tests/cxx_runtime/CMakeLists.txt:46](#)

Enclosing context (unevaluated):

```
1. if(MODE STREQUAL package) -> else()
2. if(MODE STREQUAL protected) -> elseif(MODE STREQUAL installed)
    target_compile_definitions(runtime_probe PRIVATE FOUNDATION_CONSUMER=1)
```

tests/rev_style_probe/CMakeLists.txt**cmake_minimum_required** | [tests/rev_style_probe/CMakeLists.txt:1](#)

Context: no enclosing condition/helper recorded.

```
cmake_minimum_required(VERSION 3.28)
```

project | [tests/rev_style_probe/CMakeLists.txt:2](#)

Context: no enclosing condition/helper recorded.

```
project(FoundationRevProbe LANGUAGES CXX)
```

target_include_directories | [tests/rev_style_probe/CMakeLists.txt:33](#)

Context: no enclosing condition/helper recorded.

```
target_include_directories(foundation_rev_probe PUBLIC "${FOUNDATION_REV_SOURCE}/external")
```

target_compile_definitions | [tests/rev_style_probe/CMakeLists.txt:34](#)

Context: no enclosing condition/helper recorded.

```
target_compile_definitions(foundation_rev_probe PRIVATE PROJECT_ROOT="${FOUNDATION_REV_SOURCE}")
```

set_target_properties | [tests/rev_style_probe/CMakeLists.txt:38](#)

Enclosing context (unevaluated):

1. foreach(target foundation_rev_probe foundation_rev_style)

```
set_target_properties(${target} PROPERTIES CXX_STANDARD 23 CXX_STANDARD_REQUIRED YES
CXX_EXTENSIONS NO CXX_SCAN_FOR_MODULES ON
MSVC_RUNTIME_LIBRARY "MultiThreaded$<$<CONFIG:Debug>:Debug>" )
```

target_compile_options | [tests/rev_style_probe/CMakeLists.txt:42](#)

Enclosing context (unevaluated):

1. foreach(target foundation_rev_probe foundation_rev_style)

2. if(MSVC)

```
target_compile_options(${target} PRIVATE /utf-8 /permissive-)
```

target_compile_options | [tests/rev_style_probe/CMakeLists.txt:44](#)

Enclosing context (unevaluated):

1. foreach(target foundation_rev_probe foundation_rev_style)

2. if(MSVC) -> elseif(CMAKE_CXX_COMPILER_ID MATCHES Clang)

```
target_compile_options(${target} PRIVATE -Werror=ambiguous-reversed-operator)
```

5. Presets and Cargo manifests

Every parsed manifest field is printed, including inherited preset names and symbolic paths. Preset inheritance, cache expansion, Cargo dependency selection, and target-specific conditions are not evaluated.

CMake presets: CMakePresets.json

CMakePresets.json:1

Field	Captured value
version	5
cmakeMinimumRequired.major	3
cmakeMinimumRequired.minor	24
cmakeMinimumRequired.patch	0
configurePresets[0].name	"base"
configurePresets[0].hidden	true
configurePresets[0].generator	"Ninja"
configurePresets[0].binaryDir	"\${sourceDir}/build/\${presetName}"
configurePresets[0].cacheVariables.BUILD_TESTING	"ON"
configurePresets[0].cacheVariables.CMAKE_EXPORT_COMPILE_COMMANDS	"ON"
configurePresets[1].name	"dev"
configurePresets[1].inherits	"base"
configurePresets[1].cacheVariables.CMAKE_BUILD_TYPE	"Debug"
configurePresets[2].name	"release"
configurePresets[2].inherits	"base"
configurePresets[2].cacheVariables.CMAKE_BUILD_TYPE	"Release"
configurePresets[3].name	"asan"
configurePresets[3].inherits	"dev"
configurePresets[3].cacheVariables.FOUNDATION_SANITIZERS	"ON"
buildPresets[0].name	"dev"
buildPresets[0].configurePreset	"dev"
buildPresets[1].name	"release"
buildPresets[1].configurePreset	"release"
buildPresets[2].name	"asan"
buildPresets[2].configurePreset	"asan"
testPresets[0].name	"dev"

Field	Captured value
testPresets[0].configurePreset	"dev"
testPresets[0].output.outputOnFailure	true
testPresets[0].execution.noTestsAction	"error"
testPresets[1].name	"release"
testPresets[1].configurePreset	"release"
testPresets[1].output.outputOnFailure	true
testPresets[1].execution.noTestsAction	"error"
testPresets[2].name	"asan"
testPresets[2].configurePreset	"asan"
testPresets[2].output.outputOnFailure	true
testPresets[2].execution.noTestsAction	"error"

Cargo: rust/Cargo.toml

rust/Cargo.toml:1

Field	Captured value
workspace.members[0]	"text_validation"
workspace.resolver	"2"
profile.dev.panic	"abort"
profile.release.panic	"abort"

Cargo: rust/text_validation/Cargo.toml

rust/text_validation/Cargo.toml:1

Field	Captured value
package.name	"foundation-text-validation"
package.version	"0.1.0"
package.edition	"2021"
package.rust-version	"1.63"
package.publish	false
lib.name	"foundation_rust"
lib.crate-type[0]	"staticlib"
lib.crate-type[1]	"rlib"

6. Helpers and command inventory

Helper parameters are declared names; the documentation does not execute helper calls or expand loop-generated targets.

Helper / kind	Declared parameters	Source
foundation_options function	target	CMakeLists.txt:67
foundation_notice_file function	path	cmake/BuildPolicy.cmake:40
foundation_generated_application_archive function	target producer	cmake/BuildPolicy.cmake:49
foundation_register_build_directory function		cmake/BuildPolicy.cmake:85
_foundation_capture_directory_policy function		cmake/BuildPolicy.cmake:100
_foundation_directory_targets function	directory output	cmake/BuildPolicy.cmake:225
foundation_finalize_build_policy function		cmake/BuildPolicy.cmake:240
foundation_is_generated_dependency function	path output	cmake/DependencyPaths.cmake:3
foundation_check_dependency function	kind path	cmake/DependencyPaths.cmake:23
foundation_check_dependency_values function	kind values	cmake/DependencyPaths.cmake:56
foundation_check_dependency_hash function	path expected	cmake/DependencyPaths.cmake:71
foundation_apply_runtime function	target	cmake/FoundationConfig.cmake.in:18
foundation_rust_component function	core	cmake/RustComponent.cmake:6
foundation_sdk_runtime function	target	cmake/SdkBuildRuntime.cmake:14
_foundation_sdk_install_literal function	value description	cmake/SdkBuildRuntime.cmake:57
_foundation_finalize_sdk_runtime function		cmake/SdkBuildRuntime.cmake:63
foundation_install_sdk_runtime function		cmake/SdkBuildRuntime.cmake:122
foundation_require_static_gnu_runtime function		cmake/StaticCxxRuntime.cmake:6
foundation_link_static_gnu_runtime function	target visibility	cmake/StaticCxxRuntime.cmake:27
foundation_test_prerequisites function	name	cmake/TestPrerequisites.cmake:6

Helper / kind	Declared parameters	Source
<code>_foundation_capture_test_labels</code> function		cmake/TestPrerequisites.cmake:28
<code>_foundation_collect_tests</code> function	directory output	cmake/TestPrerequisites.cmake:37
<code>foundation_finalize_test_prerequisites</code> function		cmake/TestPrerequisites.cmake:47
<code>gui_warnings</code> function	target	gui/CMakeLists.txt:111
<code>foundation_gui_executable</code> function	target source	gui/CMakeLists.txt:115
<code>foundation_gui_patch</code> function	source patch output	gui/CMakeLists.txt:145
<code>foundation_gui_repatch</code> function	source patch output	gui/CMakeLists.txt:156
<code>foundation_gui_check</code> function	name target labels	gui/CMakeLists.txt:165
<code>foundation_gui_sdl_display_test</code> function	name	gui/CMakeLists.txt:171

Complete command inventory counts

1853 captured CMake commands. Counts include control statements and declarations in every recorded branch/helper; they do not describe a single configured execution. Raw commands and source contexts are retained in `atlas.json` and the HTML explorer.

Command	Count	Printed configuration selection
<code>_foundation_capture_directory_policy</code>	1	Full inventory in HTML / JSON
<code>_foundation_collect_tests</code>	2	Full inventory in HTML / JSON
<code>_foundation_directory_targets</code>	2	Full inventory in HTML / JSON
<code>_foundation_sdk_install_literal</code>	4	Full inventory in HTML / JSON
<code>add_custom_command</code>	3	Full detail in section 4
<code>add_custom_target</code>	11	Full inventory in HTML / JSON
<code>add_dependencies</code>	9	Full detail in section 4
<code>add_executable</code>	14	Full inventory in HTML / JSON
<code>add_library</code>	12	Full inventory in HTML / JSON
<code>add_subdirectory</code>	1	Full detail in section 4
<code>add_test</code>	35	Full inventory in HTML / JSON
<code>check_cxx_source_compiles</code>	1	Full detail in section 4
<code>check_required_components</code>	1	Full inventory in HTML / JSON
<code>cmake_language</code>	5	Full inventory in HTML / JSON
<code>cmake_minimum_required</code>	4	Full detail in section 4

Command	Count	Printed configuration selection
cmake_path	4	Full inventory in HTML / JSON
configure_file	8	Full inventory in HTML / JSON
configure_package_config_file	2	Full inventory in HTML / JSON
continue	2	Full inventory in HTML / JSON
else	30	Full inventory in HTML / JSON
elseif	23	Full inventory in HTML / JSON
enable_language	1	Full detail in section 4
enable_testing	1	Full inventory in HTML / JSON
endforeach	69	Full inventory in HTML / JSON
endfunction	29	Full inventory in HTML / JSON
endif	255	Full inventory in HTML / JSON
endwhile	1	Full inventory in HTML / JSON
execute_process	17	Full inventory in HTML / JSON
file	44	Full inventory in HTML / JSON
find_package	8	Full detail in section 4
find_program	6	Full detail in section 4
foreach	69	Full inventory in HTML / JSON
foundation_apply_runtime	2	Full inventory in HTML / JSON
foundation_check_dependency	3	Full inventory in HTML / JSON
foundation_check_dependency_values	1	Full inventory in HTML / JSON
foundation_generated_application_archive	1	Full inventory in HTML / JSON
foundation_gui_check	16	Full inventory in HTML / JSON
foundation_gui_executable	18	Full inventory in HTML / JSON
foundation_gui_patch	18	Full inventory in HTML / JSON
foundation_gui_repatch	6	Full inventory in HTML / JSON
foundation_gui_sdl_display_test	2	Full inventory in HTML / JSON
foundation_install_sdk_runtime	1	Full inventory in HTML / JSON
foundation_is_generated_dependency	4	Full inventory in HTML / JSON
foundation_link_static_gnu_runtime	4	Full inventory in HTML / JSON
foundation_notice_file	2	Full inventory in HTML / JSON
foundation_options	14	Full inventory in HTML / JSON
foundation_register_build_directory	2	Full inventory in HTML / JSON
foundation_require_static_gnu_runtime	4	Full inventory in HTML / JSON
foundation_rust_component	1	Full inventory in HTML / JSON

Command	Count	Printed configuration selection
foundation_sdk_runtime	1	Full inventory in HTML / JSON
foundation_test_prerequisites	24	Full inventory in HTML / JSON
function	29	Full inventory in HTML / JSON
get_filename_component	10	Full inventory in HTML / JSON
get_property	31	Full inventory in HTML / JSON
get_target_property	24	Full inventory in HTML / JSON
if	255	Full inventory in HTML / JSON
include	19	Full detail in section 4
include_directories	2	Full detail in section 4
include_guard	7	Full inventory in HTML / JSON
install	33	Full inventory in HTML / JSON
list	56	Full inventory in HTML / JSON
math	5	Full inventory in HTML / JSON
message	84	Full inventory in HTML / JSON
option	10	Full inventory in HTML / JSON
project	4	Full detail in section 4
return	13	Full inventory in HTML / JSON
set	208	Full inventory in HTML / JSON
set_and_check	1	Full inventory in HTML / JSON
set_property	37	Full detail in section 4
set_target_properties	13	Full detail in section 4
set_tests_properties	34	Full inventory in HTML / JSON
string	81	Full inventory in HTML / JSON
target_compile_definitions	9	Full detail in section 4
target_compile_features	7	Full detail in section 4
target_compile_options	15	Full detail in section 4
target_include_directories	17	Full detail in section 4
target_link_libraries	29	Full inventory in HTML / JSON
target_link_options	14	Full detail in section 4
target_sources	3	Full inventory in HTML / JSON
unset	3	Full inventory in HTML / JSON
while	1	Full inventory in HTML / JSON
write_basic_package_version_file	1	Full inventory in HTML / JSON

[Complete build inventory](#) | All captured command records

7. Recorded limitations and warnings

1. cmake/RustComponent.cmake:173: target_link_libraries references \${core}, whose declaration is absent or supplied elsewhere
2. CMake is mapped lexically, never executed: targets show a union of declarations, not one selected configuration. Conditions and helper/loop contexts remain explicit.
3. Variables, environment/cache references and generator expressions are unresolved; symbolic source and link entries are preserved verbatim.
4. Includes, subdirectories, globbing, package discovery, generated files, retained-input restoration and external commands are not evaluated or invoked.
5. Conditional providers/backends, loops and function bodies may describe mutually exclusive or template targets. Source and link statements retain their separate conditions.

This PDF is a printable view of captured evidence. File registration, compiler flags, conditions, and provider ownership must be checked in the linked source before editing. The retained GUI supplier references, where included elsewhere in this snapshot, are unpatched archive members and do not represent a configured dependency.

[Up: What to edit](#) | [Compiler explorer](#)